1	**C** density of ruler = 13.78 /(31.2 × 0.32) = 1.3802 g cm^{-3} fractional uncertainty of density = $\frac{0.001}{0.312} + \frac{0.02}{0.32} + \frac{0.01}{13.78} = 0.066$ percentage uncertainty of density = 0.066 × 100% = 6.6 % actual uncertainty of density = 0.066 × 1.3802 = 0.09 g cm^{-3} Hence, density = 1.38 ± 0.09 g cm^{-3}
2	**B**